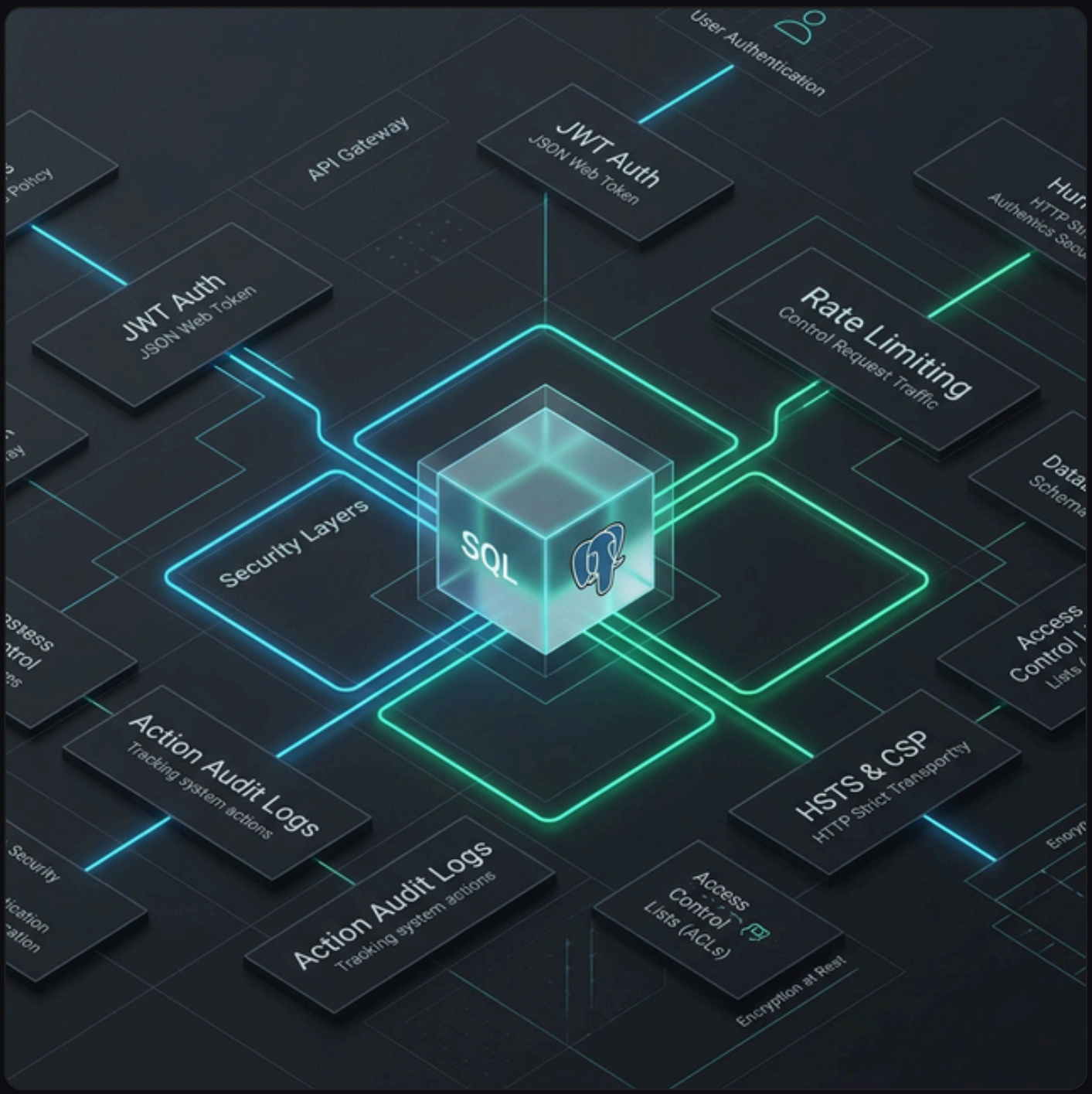
Security & Data Integrity Architecture

BUILDING SECURE CLOUD APPLICATIONS BY DEFAULT



When I build a backend, security isn't an afterthought. It's the foundation. Here's a technical breakdown of the protection layers I implement in every FastAPI/PostgreSQL project.

> 1. Access Control

JWT (JSON Web Tokens)

Stateless authentication with custom expiration logic. Prevents session hijacking and ensures secure communication between the frontend and the API.

Rate Limiting

Automatic protection against Brute-Force attacks. My APIs automatically block suspicious IP behaviors to prevent unauthorized access attempts.

> 2. Data Accountability

Automated Action Audit Logs

I implement a full traceability system. Every 'Create', 'Update', or 'Delete' action is recorded with a timestamp, IP, and User ID. Never lose track of what happened in your database.

```
# Sample Audit Entry Structure
{ "action": "UPDATE_ORDER", "user": "admin@user.com", "timestamp": "2024-03-04T10:08", "status": "SUCCESS" }
```

> 3. Database Health

Relational Integrity

Using PostGIS for complex location-based data and SQLAlchemy for ORM mapping, ensuring data consistency across all tables.

Automated Migrations

Full control over database versions using Alembic. This allows for zero-downtime updates and easy data restoration.

Built with Secure Development Lifecycle (SDL) Principles.